Bago National High School

General Mathematics (Grade 11) · Term 2 Week 9 Student Handout

November 9–13, 2026 · Teacher: Raymart Valbarez

Name: _____ Section: _____ Date: _____

Light, self-paced ADM activity (study indoors; rest if tired or hot).

How to use this handout

1. Set your scientific calculator to DEG (degree mode), not RAD.
2. Study at your own pace. This is light home study — rest if you feel tired or hot; stay indoors. No outdoor measuring is required.
3. Write your Practice answers in a notebook. You do not need to print or take photos of this file.

Lesson 1 — Bearings (N θ E)

A bearing tells direction from north. "N 40° E" means: face north, then turn 40° toward east. Same idea for S θ W, N θ W, and S θ E. For a two-leg trip that makes a right triangle, find the straight-line distance with Pythagoras, then the angle with $\tan^{-1}$.

$d = \sqrt{(x^2 + y^2)}$ · $\theta = \tan^{-1}$(E–W leg ÷ N–S leg) · write as N θ E (or S θ W, …)

Worked example (Culasi Port banca)

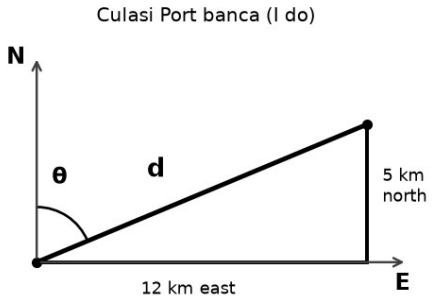
A banca sails 12 km east, then 5 km north from Culasi Port.

Step 1: Sketch east and north legs (right triangle).

Step 2: Distance $d = \sqrt{(12^2 + 5^2)} = \sqrt{(144 + 25)} = \sqrt{169} = 13$ km.

Step 3: Angle from north $\theta = \tan^{-1}(12 \div 5) = \tan^{-1}(2.4) = 67.4°$.

Step 4: Bearing from the port = N 67.4° E.



Lesson 2 — Law of Sines (AAS / ASA)

Use the Law of Sines when the triangle is not right-angled and you know two angles and one side. Side a is opposite angle A. First find the third angle (angles add to 180°), then pair each known side with its opposite angle.

$a / \sin A = b / \sin B = c / \sin C$

Worked example (triangular lot)

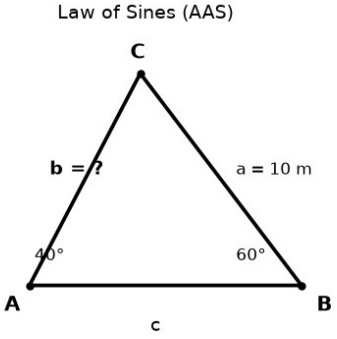
Given: A = 40°, B = 60°, a = 10 m. Find side b.

Step 1: C = 180° − 40° − 60° = 80°.

Step 2: $b / \sin 60° = 10 / \sin 40°$.

Step 3: $b = 10 \times \sin 60° \div \sin 40° = 13.47$ m.

(Round sides to 2 decimal places.)



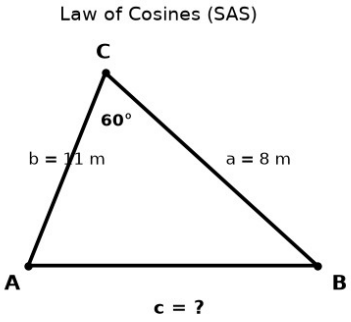
Lesson 3 — Law of Cosines (SAS / SSS)

Use the Law of Cosines when you know two sides and the included angle (SAS) — find the third side — or when you know three sides (SSS) — find an angle. Angle C is the angle between sides a and b; side c is opposite C.

$c^2 = a^2 + b^2 - 2ab\cos C$ • $\cos C = (a^2 + b^2 - c^2) \div (2ab)$, then $C = \cos^{-1}(\ldots)$

Worked example (Brgy. Bago fishpond / lot)

Given: sides 8 m and 11 m, included angle 60°. Find c.
Step 1: $c^2 = 8^2 + 11^2 - 2(8)(11)\cos 60°$.
Step 2: $\cos 60° = 0.5 \rightarrow c^2 = 64 + 121 - 88 = 97$.
Step 3: $c = \sqrt{97} = 9.85$ m.



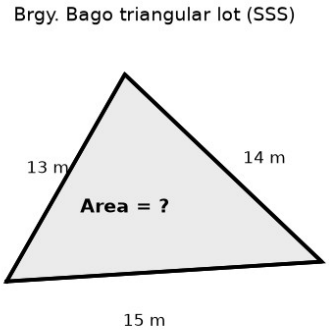
Lesson 4 — Area of oblique triangles

If you know two sides and the included angle (SAS), use Area = ½ab sin C. If you know three sides only (SSS), use Heron's Formula: find s first (half the perimeter), then take the square root. Always write square units (m², cm², dm²).

SAS: Area = ½ ab sin C • SSS (Heron): s = (a+b+c)/2 ; Area = √[s(s−a)(s−b)(s−c)]

Worked examples

SAS: Area = ½(10)(12) sin 30° = 60 × 0.5 = 30 m².
SSS lot 13 m, 14 m, 15 m:
s = (13+14+15)/2 = 21
Area = √(21·8·7·6) = √7056 = 84 m².



Which tool? (quick chooser)

If you have…	Use…
Right triangle (includes 90°)	SOH-CAH-TOA
Two angles + one side (AAS/ASA)	Law of Sines
Two sides + included angle (SAS), or three sides (SSS) for a side/angle	Law of Cosines
Two sides + included angle, need area	Area = ½ab sin C
Three sides, need area	Heron's Formula

Practice (write answers in your notebook)

Round sides to 2 decimal places and angles to 1 decimal place. Calculator in DEG mode. No outdoor measuring.

1. (Bearings) A banca leaves Culasi Port, sails 7 km north, then 24 km east. Find (a) the straight-line distance from the port and (b) the bearing of its position from the port in N θ E form.

2. (Bearings) A boat sails 10 km south, then 10 km west. Find the straight-line distance from the start and write the bearing in S θ W form.

3. (Law of Sines) In △ABC, A = 25°, B = 55°, and side a = 12 m. Find angle C and side b.

4. (Law of Sines) In △ABC, A = 42°, C = 68°, and side c = 18 m. Find side a.

5. (Law of Cosines) A Brgy. Bago fishpond corner has sides 9 m and 14 m with included angle 55°. Find the third side c.

6. (Law of Cosines) A triangular lot has sides 8 m, 10 m, and 13 m. Find the angle opposite the 10 m side.

7. (Area) A triangular sail has sides 11 dm and 16 dm with included angle 35°. Find its area.

8. (Area — Heron) A garden lot in Brgy. Bago has sides 11 m, 13 m, and 16 m. Find its area using Heron's Formula.

9. (Which tool?) You know two angles and one side of a non-right triangle, and you need another side. Which tool do you use?

10. (Which tool?) You know three sides of a triangular lot and need its area. Which formula do you use?

Common mistakes to avoid

- DEG vs RAD — always check DEG mode before pressing sin, cos, tan, or their inverses.
- Opposite vs adjacent — side a is across from angle A (not next to it).
- Label carefully — in the Law of Cosines, C is the angle BETWEEN sides a and b.
- Units for area — write m^2 (or cm^2 / dm^2), not plain meters.

Rounding reminder: sides → 2 d.p.; angles → 1 d.p.